CHAPTER 8

MOMENTS SKEWNESS AND KURTOSIS

CHAPTER ESSENCE AT A GLANCE:
Master raw and central moments, Karl Pearson's and Bowley's measures of skewness, and Pearsonian kurtosis classification (Leptokurtic, Mesokurtic, Platykurtic).

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Raw Moments vs Central Moments:** Moments about arbitrary origin A are raw moments; moments about arithmetic mean are central moments (mu1 = 0, mu2 = variance, mu3 measures skewness, mu4 measures kurtosis).

• **Concept of Skewness (Asymmetry):** Measures lack of symmetry in a distribution; symmetric distribution has Mean = Median = Mode and Sk = 0; right-skewed has positive tail (Mean > Median > Mode); left-skewed has negative tail.

• **Karl Pearson's and Bowley's Coefficients:** Karl Pearson's Sk = (Mean - Mode) divided by SD = 3(Mean - Median) divided by SD; Bowley's Sk_B = (Q3 + Q1 - 2M) divided by (Q3 - Q1), bounded between -1 and +1.

• **Kurtosis and Peakedness:** Kurtosis measures degree of peakedness or flatness relative to the normal curve; measured by Beta2 = mu4 divided by $(mu2)^2$; Mesokurtic (Beta2 = 3), Leptokurtic (Beta2 > 3), Platykurtic (Beta2 < 3).

ESSENTIAL STATISTICAL FORMULAS & MEASURES

$$\text{Central Moments} \iff \mu_1 = 0, \quad \mu_2 = \sigma^2, \quad \mu_3 = \mu_3' - 3\mu_2'\mu_1' + 2(\mu_1')^3$$

Fundamental Relationships Linking Raw to Central Moments

$$\text{Karl Pearson Skewness} \iff \text{Sk}_p = \frac{\bar{x} - \text{Mode}}{\sigma} = \frac{3(\bar{x} - \text{Median})}{\sigma}$$

Standardized Measure of Distributional Asymmetry

$$\text{Bowley Quartile Skewness} \iff \text{Sk}_B = \frac{(Q_3 - M) - (M - Q_1)}{Q_3 - Q_1} = \frac{Q_3 + Q_1 - 2M}{Q_3 - Q_1}$$

Positional Measure of Skewness Bounded in [-1, +1]

$$\text{Kurtosis} \iff \beta_2 = \frac{\mu_4}{\mu_2^2} \to \{ > 3 : \text{Leptokurtic}, \quad = 3 : \text{Mesokurtic}, \quad < 3 : \text{Platykurtic} \}$$

Pearsonian Fourth-Moment Peakedness Classification

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** The first central moment mu1 about arithmetic mean is ALWAYS identically zero; if your calculation yields a non-zero value, check arithmetic immediately.

• **Alert:** For a perfectly symmetric distribution, Mean = Median = Mode, and both Karl Pearson's and Bowley's coefficients equal zero.

• **Alert:** In formula card 4, ensure Beta2 is evaluated relative to 3 (Gamma2 = Beta2 - 3); a normal distribution has Gamma2 = 0.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Define Central Moments. What are the values of the first central moment (mu1) and second central moment (mu2)? [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Definition:** The r-th central moment (mu_r) is the arithmetic mean of the r-th powers of deviations of observations taken from their arithmetic mean. **[1 Mark]**

• **Values:** mu1 is always identically zero (mu1 = 0); mu2 is equal to the variance of the distribution (mu2 = σ^2). **[1 Mark]**

2. For a frequency distribution, Mean = 60, Median = 56, and Standard Deviation = 12. Calculate Karl Pearson's coefficient of skewness and state the nature of symmetry. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **Formula:** Sk_p = 3 * (Mean - Median) divided by Standard Deviation. **[1.5 Marks]**

• **Calculation:** Sk_p = 3 * (60 - 56) divided by 12 = 3 × 4 divided by 12 = 12 divided by 12 = +1.0. **[1.5 Marks]**

• **Nature of Symmetry:** Since Sk_p is positive (+1.0), the distribution is positively skewed with a longer tail stretched to the right side (Mean > Median). **[1 Mark]**

3. In a distribution, Lower Quartile Q1 = 20, Median = 30, and Upper Quartile Q3 = 50. Calculate Bowley's coefficient of skewness. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Formula:** Sk_B = (Q3 + Q1 - 2 * Median) divided by (Q3 - Q1). **[1 Mark]**

• **Calculation:** Sk_B = (50 + 20 - 2 × 30) divided by (50 - 20) = (70 - 60) divided by 30 = 10 divided by 30 = +0.33. **[1.5 Marks]**

• **Interpretation:** Moderately positively skewed distribution. **[0.5 Mark]**

4. What is Kurtosis? Distinguish between Leptokurtic, Mesokurtic, and Platykurtic distributions using Beta2. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Definition:** Kurtosis measures the degree of peakedness or flatness of a frequency distribution curve relative to a normal distribution curve. **[1 Mark]**

• **Three Types:**
• **Mesokurtic:** Normal peakedness curve with Beta2 = 3.
• **Leptokurtic:** More peaked curve with fatter tails and Beta2 > 3. **[1 Mark]**

• • **Platykurtic:** Flatter topped curve with thinner tails and Beta2 < 3. **[1 Mark]**

5. State the relationship between Mean, Median, and Mode in: (a) Symmetrical distribution, (b) Positively skewed distribution. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **(a) Symmetrical:** Mean = Median = Mode (all three measures coincide at the center). **[1 Mark]**

• **(b) Positively Skewed:** Mean > Median > Mode (Mean is pulled furthest towards the right tail). **[1 Mark]**

6. Write the algebraic formula expressing the third central moment (mu3) in terms of raw moments about the origin. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Formula:** mu3 = mu3' - 3 * mu2' * mu1' + 2 * $(mu1')^3$, where mu1', mu2', and mu3' are the first three raw moments about origin zero. **[2 Marks]**